



Infrastructure Without Utilization? A Transport System Policy and Governance Analysis of Terminal Performance and Public Transport Integration in Padang City

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Received: 11-11-2025

Revised: 01-19-2026

Accepted: 01-26-2026

Citation: Z. Alhadi, A. H. R., R. Yusran, and F. Eriyanti, "Infrastructure without utilization? A transport system policy and governance analysis of terminal performance and public transport integration in Padang City," *Int. J. Transp. Dev. Integr.*, vol. 10, no. 1, pp. 219–246, 2026. <https://doi.org/10.56578/ijtdi100116>.



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Abstract: Public transport terminals are expected to operate as functional nodes within urban transport systems, facilitating transfers and supporting network efficiency. However, in many cases, newly developed terminals remain underutilized despite meeting infrastructure standards. This study investigates such a situation in Padang City, Indonesia, focusing on a Type A terminal that has not achieved its intended operational role. The analysis is based on a qualitative case study combining field observations, interviews with terminal managers, operators, and users, and a review of regulatory and operational documents. Rather than examining infrastructure conditions alone, the study looks at how the terminal is positioned within the wider transport system and how institutional arrangements influence its use in practice. The results indicate that low utilization is closely linked to weak system integration. In particular, limited last-mile access, the absence of reliable feeder services, and mismatched operating schedules reduce the practicality of using the terminal. These conditions affect both passengers and operators, making alternative departure points more attractive. At the same time, fragmented responsibilities between different levels of government reduce the consistency of implementation and enforcement, which further discourages compliance with terminal-based operations. Taken together, these factors create a situation in which the terminal functions below its intended capacity. Improving performance therefore requires more than infrastructure provision. Greater attention needs to be given to network integration, coordination between responsible agencies, and the alignment of operational practices with system-level objectives. This case study highlights the importance of viewing terminal governance performance as part of broader transportation system policy, rather than as an isolated facility.

Keywords: Transportation infrastructure; Transportation policy; Terminal governance; Effectiveness; Coordination; Padang City

1 Introduction

Public transportation is a crucial component supporting public mobility, economic efficiency, and environmental sustainability in urban areas [1–5]. A crucial element of the public transportation system is the existence of terminals, which function as distribution hubs for passenger and vehicle movements [6–9]. The management of terminals as public transportation hubs is regulated by Law Number 22 of 2009 concerning Road Traffic and Transportation, which stipulates that passenger terminals are part of the transportation infrastructure that must be managed safely, orderly, and organized. Terminal management is regulated by Regulation of the Minister of Transportation of the Republic of Indonesia Number PM 132 of 2015 concerning the Management of Road Passenger Terminals, which states that a terminal is a base for public motorized vehicles used to manage arrivals and departures, pick up and drop off passengers and/or goods, and transfer between modes of transportation (Minister of Transportation Regulation No. 132 of 2015). The existence of terminals in an area triggers economic activity in the surrounding area. The importance of transportation is evident in the increasing demand for mobility services for people and goods throughout the region [10–14].

In Padang City, West Sumatra Province, the growing demand for public transportation poses a unique challenge

amid the city's rapid population growth and socio-economic activities. Padang City Regional Regulation Number 2 of 2022 regulates the implementation of land transportation, including terminals, as a vital part of the transportation system. This regulation aims to improve the economy and public welfare by providing orderly, safe, and comfortable transportation. According to data from the Central Statistics Agency, the population of Padang City in 2024 was 954,177. Padang City covers an area of 694.96 km². This high population requires an adequate, integrated public transportation system that effectively serves the community's mobility needs.

Anak Air Terminal is a Type A passenger terminal still active in Padang City. It is located on Anak Air, Batipuh Panjang Village, Koto Tengah District, Padang City. Anak Air Terminal began operations in October 2021, with a budget of over Indonesian Rupiah (IDR) 90 billion. The terminal covers an area of 27,385 m² and a building area of 10,364 m². It is under the authority of the Ministry of Transportation of the Republic of Indonesia through the Land Transportation Management Agency for the West Sumatra region. Anak Air Terminal is designated to serve various types of public transportation, including:

(a) Inter-City and Inter-Provincial Transportation: This is a public transportation service that transports passengers from one city to another within a different province. Inter-City and Inter-Provincial Transportation is a long-distance transportation service and typically uses large buses. For example, the ANS bus operates the Padang-Jakarta-Bandung route.

(b) Intra-Provincial City Transportation: A public transportation service that transports passengers between cities within the same province. Intra-Provincial City Transportation typically uses medium-sized buses or minibuses.

(c) City Transportation: A type of local transportation that serves mobility within the city, with short routes and a small capacity.

(d) Trans Padang (Bus Rapid Transit): This is an urban mass transportation system that serves as the backbone of public transportation in Padang City. The Anak Air Terminal is served by two main Trans Padang corridors: Corridor 1 (Pasar Raya-Anak Air Terminal) and Corridor 4 (Anak Air Terminal-Teluk Bayur).

Although the Anak Air Terminal has been designed to accommodate various modes of transportation, field observations and data collection indicate that its operational effectiveness is not yet evident. Various facilities and infrastructure have been provided, such as passenger waiting rooms, ticket counters, vehicle parking, closed-circuit television (CCTV), public/disabled restrooms, maternity/nursing rooms, prayer rooms, a canteen, crew rest areas, a minor vehicle repair shop, and a micro, small, and medium enterprises (MSME) area. However, these facilities have not been optimally utilized by transportation operators or the public using the services. One particularly attractive feature of the Anak Air Terminal is the multipurpose room, which can be used for social, educational, or community activities. According to information from terminal management, this room has previously been used by elementary schools or kindergartens to host arts performances. This demonstrates that the terminal plays a role not only in transportation but also has the potential to serve as a multifunctional public space that can strengthen social ties and activities within the surrounding community. However, this potential has not been fully utilized due to low passenger and fleet numbers. Therefore, the utilization of existing facilities has not had a significant impact on the economic and social activities that should be generated within the terminal area.

This situation is inseparable from the inhibiting factors in the terminal's structural governance, particularly related to location and accessibility. This location is far from the city center and major public activities. The alternative routes used by large buses are also relatively narrow, limiting movement in both directions, causing local congestion, especially during rush hour. Furthermore, the terminal's access road is also used by private vehicles and residents, contributing to traffic congestion in the area. The lack of supporting infrastructure, such as pedestrian sidewalks, and the absence of dedicated public transportation lanes exacerbate these conditions. Furthermore, access to the terminal is poorly supported by traffic signs or very limited directional signs. These limitations discourage public transportation operators from using the official terminal. This discrepancy demonstrates a mismatch in perception between terminal managers and transportation operators, reflecting weak communication, coordination, and stakeholder engagement. This is a key factor hampering the effectiveness of terminal governance. This phenomenon also highlights that terminal governance is not yet optimal. If not addressed seriously, these early stages could continue to fuel the existence of illegal terminals and deviate official terminals from their ideal function as the primary hub for orderly, safe, and efficient public transportation.

This governance encompasses crucial aspects such as the formulation of management policies, inter-agency coordination, supporting infrastructure, public services, and enforcement of regulations. Unfortunately, the proliferation of illegal terminals reflects the inability of official terminals to meet user needs and expectations. Consequently, oversight and enforcement by the Transportation Agency are difficult, negatively impacting public trust in the public transportation system. Before the Anak Air terminal was inaugurated and put into operation, Padang City had been without a bus terminal for 12 years. This was due to the Bingkuang regional terminal, located on the Bypass Km 15 in Aia Pacah Village, not functioning optimally. This was due to the terminal's location on the outskirts of the city, far from the city's service centers, the lack of supporting facilities and infrastructure, and the emergence of alternative transportation in the form of travel and illegal terminals. Consequently, the terminal ceased

operations in 2007. Furthermore, the 2009 earthquake that struck Padang City damaged the Padang Government infrastructure. Consequently, the terminal was converted into Padang City Hall.

Table 1. Bus number data

No.	Bus Operating Company	Bus Services	Bus Destination	Departure Schedule	Nominal Rate (IDR)
1	Putra Pelangi	Executive Class	Medan	10.15	310.000
			Banda Aceh	10.15	600.000
		Suite Combi Class	Jakarta/Bogor	08.30	875.000
2	Sembodo	Suite Class	Jakarta/Bogor	08.30	900.000
			Jakarta/Bogor	08.30	775.000
		KGC KGC	Jakarta	10.15	650.000
3	Transport Express Jaya		Bandung/Sukabumi Jakarata	10.15	680.000
		Executive Class	Bandung/Sukabumi	10.15	475.000
				10.15	500.000
		S.F Sleeper	Bekasi Bekasi	09.15	750.000
4	Al Hijrah	S.F Sleeper Royal Platinum Class	Bogor	09.15	775.000
				09.15	675.000
5	Epa Star	Executive Class	Palembang	09.00	325.000
6	F amiliy Raya Ceria	Executive Class	Solo	10.30	650.000
		Executive Plus	Jambi	16.25	230.000
7	Palala	Panorama Class	Jakarta/Jabodetabek	09.00	670.000
		Platinum Class	Jakarta/Karawang	09.00	700.000
		Platinum Pluss	Jakarta/Karawang	09.00	770.000
8	Miyor Prima Abadi	Ultimate doble	Jakarta/Karawang	09.00	850.000
		Ultimate single	Jakarta/Karawang	09.00	870.000
		Dream class	Jakarta/Karawang	09.00	950.000
9	SAN	Executive Class	Bengkulu	11.45	300.000
		Executive Class	Palembang Lampung	09.30	500.000
			Jabodetabek Palembang Lampung	09.30	500.000
				09.30	500.000
			Jabodetabek	09.30	675.000
				09.30	675.000
				09.30	675.000
			Kisaran Medan Binjai	09.00	265.000
			Kisaran Medan Binjai		285.000
			Duri	09.00	300.000
10	NPM	Sultan Class	Dumai Muaro Tebo	09.00	350.000
			Muaro	10.30	300.000
				10.30	350.000
				17.00	375.000
				17.00	375.000
			Bulian/Jambi	17.30	200.000
					225.000
				17.30	175.000
					215.000
11	MPM	Super Executive Class	Jakarta	09.30	625.000
		Premium Class	Jakarta	09.30	650.000
		Executive Royal Class	Jakarta/Bandung	10.00	550.000
12	ANS	Luxury Class	Jakarta/Bandung	10.00	650.000
					700.000

However, in practice, the Anak Air Terminal has not operated optimally. Despite adequate facilities, utilization remains very low. Based on field data, 12 intercity and intercity buses were registered at the terminal in 2024, and in January 2025, the addition of Damri tourist buses served routes including Padang-Bukittinggi, Padang-Maninjau, Padang-Painan, and Muaro Lasak-Bukittinggi. Data on buses operating at the Anak Air terminal can be seen in Table 1.

Data from the various major POs listed as official operators indicates that the single-departure daily departure schedule, coupled with the dominance of Inter-City and Inter-Provincial Transportation routes, indicates that the function of air terminals as hubs for daily and regional public mobility is not yet fully operational. Furthermore, most POs maintain branch offices or independent pools in city centers, opting not to rely entirely on the official terminal

system, citing affordability and consumer preference. Meanwhile, Intra-Provincial City Transportation is virtually inactive at terminals, even though this segment should serve as the basis for interregional transportation within the province and is crucial for supporting local community mobility. These facts emphasize that the physical presence of terminals and a diversity of services alone are insufficient to ensure effective terminal governance. The continued lack of synergy between terminal managers, transport operators, and the community points to deep-seated issues in institutional aspects, oversight, outreach, and the integration of transportation policies.

The urgency of this research lies in the need for a comprehensive evaluation and mapping of the various factors hindering optimal terminal management. This research seeks to address a current research gap: the lack of in-depth and specific academic studies on the management of Type A terminals, particularly in West Sumatra. Previous research has tended to focus on technical aspects of facilities or user satisfaction, without linking them to local institutions and contexts. This research is also important because improving terminal governance not only impacts the quality of public transportation services but also supports the creation of environmentally friendly, sustainable, and spatially organized cities. Furthermore, it can serve as a reference for policymakers in formulating strategies for optimizing terminal management that adapt to community needs and local challenges. This study is expected to provide data-based recommendations on strengthening institutional coordination, developing supporting infrastructure, digitizing service information, and empowering micro-economic actors around the terminal. This way, terminals will not only be physical structures but also become vibrant, beneficial transportation hubs.

This paper primarily focuses on transport governance and public service effectiveness, with terminals treated not simply as physical assets but as regulated service nodes within an urban mobility system. While the study substantively addresses transport planning and infrastructure management, its core contribution is explaining why public transport infrastructure can remain underutilized despite regulatory compliance and adequate facilities, a fundamentally institutional and implementation-oriented question. Therefore, the analysis highlights multi-actor coordination, enforcement credibility, and operator behavior as mechanisms through which policy intent translates into service utilization. In this regard, the case of Anak Air Terminal is used as an empirical lens to contribute to the broader debate on implementation failures, institutional fragmentation, and the conditions under which public transport facilities produce effective service outcomes, rather than as a purely technical assessment of terminal design or capacity.

To strengthen this paper's position within transportation systems research, this study treats Anak Air Terminal not only as a regulated public facility but also as a network node whose performance can be understood through fundamental transportation system concepts. In transportation engineering terms, terminal effectiveness is reflected in how well the node converts its designed capacity into actual flows and reliable intermodal transfers, as measured through simple operational indicators such as average daily departures and arrivals, passenger volumes, transfer feasibility, and route/connector connectivity. Therefore, this analysis interprets underutilization as a system performance issue, rather than simply an administrative deficiency: limited node accessibility and weak connection integration increase overall travel costs, reduce service level reliability, and weaken the terminal's attractiveness compared to informal curbside pickup points. By combining these system performance concepts with governance mechanisms such as coordination, enforcement credibility, and operator incentives, this study explains why an infrastructure-led approach can result in a technically complete but functionally weak terminal as an intermodal node in an urban mobility network.

2 Literature Review

2.1 Concept of Effectiveness

Effectiveness is the consciously predetermined utilization of resources, facilities, and infrastructure to produce a specific quantity of goods or services within the activities undertaken [15–18]. Effectiveness also measures success based on the achievement of predetermined goals. The closer the results are to the set targets, the higher the level of effectiveness. Siddiquei et al. [16] defined work effectiveness as the timely completion of work as previously determined. Effectiveness indicates an organization's ability to accurately achieve its established goals [19–22]. Achieving established targets, applicable metrics, and standards reflects an organization's commitment to operational effectiveness [23–25].

The factors influencing effectiveness, as proposed by Steers [26] in his book *Organizational Effectiveness*, include: First, organizational characteristics. Organizations managing terminals and public transportation must have a flexible yet coordinated structure and an efficient bureaucratic system. Second, environmental characteristics. External conditions greatly influence the effectiveness of transportation management. Third, employee characteristics. The human resources (HR) directly involved in public services play a crucial role in service effectiveness. In fact, organizational employees are the most influential factor in effectiveness because it is employee behavior that, in the long term, facilitates the achievement of organizational goals. Fourth, management characteristics, strategies, and work mechanisms are designed to condition the entire organization to achieve effectiveness.

2.2 Relevant Studies

Weicker [27] discussed the detrimental impact of poorly conceived transport modernisation policies in Volgograd, highlighting how the replacement of a minibus system with a bus network failed due to the exclusion of local stakeholders from decision-making. This governance issue illustrates that infrastructure, such as bus terminals, can become ineffective when not aligned with the needs and voices of the public. The case emphasizes the importance of local knowledge and cooperation in ensuring effective public transport systems.

Peters [28] discussed how public transport in Surabaya emerges from the limits of heavy infrastructure, where traditional governance fails. It highlights the improvisation of local residents who repurpose remnants of industrial infrastructure to create effective public transport solutions, such as becak and bemo. This grassroots approach demonstrates that infrastructure can exist without effective utilization by formal governance, as communities adapt and innovate to meet their transport needs, thereby asserting their right to infrastructure in a post-colonial context.

Calderón-Ramírez et al. [29] highlighted a significant misalignment between Mexicali's public transport network and the commuting needs of its population, particularly in marginalized areas. It emphasizes that traditional transportation planning has favored private car infrastructure, leading to inefficiencies in public transport. Users experience long travel times, low service frequency, and limited connectivity, which undermines the effectiveness of public transport systems. The research advocates for geographic information system (GIS)-based spatial analysis to redesign public transport and enhance urban mobility governance towards sustainability and social inclusion.

Cirolia and Harber [30] explored the governance of transport infrastructures, particularly in sub-Saharan African cities, highlighting how urban statecraft influences the effectiveness of public transport systems, including bus terminals. It examines the dynamics of infrastructure utilization, emphasizing that effective governance is crucial for ensuring that transport facilities serve their intended purpose. The analysis reveals that without proper governance and operational strategies, infrastructure like bus terminals may remain underutilized, impacting overall public transport effectiveness in urban settings.

Jnr [31] discussed the importance of governance in enhancing public transportation effectiveness, particularly through the sustainable mobility governance model. It emphasizes the need for integrated data-driven services to improve the utilization of infrastructure, such as bus terminals, by promoting accessibility and multimodality. The findings suggest that addressing both technical and non-technical factors is crucial for optimizing public transport systems, thereby ensuring that infrastructure is effectively utilized to support sustainable urban mobility and improve overall commuter behavior.

While previous studies often discuss governance, infrastructure, finance, security, and user preferences as determinants of terminal performance in parallel, this paper adopts a more explicit mechanism-based structure. Specifically, the paper posits a key causal logic where institutional coordination shapes implementation strength. Stronger implementation, in turn, increases operator participation and compliance because the perceived benefits of complying with terminal-based operations outweigh the costs of non-compliance. Higher operator participation then increases terminal utilization, which ultimately determines governance effectiveness, reflected in orderly operations, service integration, safety, and policy legitimacy. Within this framework, variables such as physical infrastructure, financial incentives, security conditions, and mobility culture are not neglected; rather, they are positioned as moderators that condition the size and direction of effects at specific stages in the chain.

This section reviews the literature thematically and problem-based, rather than as a catalog of sources. It is structured around three debates that directly inform the research problem and analytical logic of this manuscript: (1) why infrastructure provision does not automatically translate into utilization; (2) how institutional fragmentation produces coordination failures that undermine implementation; and (3) why mismatches between formal policy design and user/operator behavior undermine performance. Each subsection concludes by identifying an unresolved issue that this study addresses empirically. In this study, development is defined as a process aimed at improving the capacity and functionality of public transport services through a combination of (i) physical provision and (ii) institutional-operational arrangements. In contrast, performance refers to the observable outcomes of such development, as reflected in how terminals and their associated services actually function in practice, as measured by service outputs and user-relevant outcomes. To avoid conceptual overlap, this analysis distinguishes several performance dimensions: effectiveness, efficiency, service coverage, equity, and accessibility. This conceptual separation makes it clear that improvements or optimizations in terminal governance cannot be treated as mere normative claims; rather, they must be grounded in specific performance dimensions and indicators aligned with empirical evidence.

A recurring debate in transportation and public infrastructure studies concerns the assumption that providing facilities will generate use and improve service outcomes. Empirical research on terminals, stations, and public transportation hubs shows that utilization depends not only on physical availability but also on service reliability, accessibility, perceived convenience, and alignment of operational incentives. In many situations, investments in terminals result in underutilized infrastructure when the service ecosystem fails to adapt, with routes bypassing terminals, passengers continuing to prefer curbside pickup, and operators avoiding waiting-time charges. This line

of research emphasizes that utilization is a behavioral and institutional outcome, not a direct function of the assets built. While previous studies have documented the gap between infrastructure and its utilization, they often provide limited explanations of the institutional and behavioral mechanisms that translate infrastructure provision into (non-)use. This study addresses this gap by examining how coordination and implementation conditions shape operator participation and, ultimately, terminal utilization.

A second body of literature highlights that effective governance in public transportation systems depends on coherent institutional arrangements. Where mandates overlap and responsibilities are distributed across multiple agencies, coordination problems arise: agencies may pursue divergent priorities, enforcement routines may be inconsistent, and shared operational protocols may be absent. Studies of multi-actor service delivery illustrate how fragmented authority can result in weak implementation, discretionary enforcement, and limited accountability, especially when coordination mechanisms are absent. Specifically in transportation governance, coordination failures can prevent terminals from serving as central organizing nodes for routing, scheduling, and enforcement. Although coordination failures are widely recognized, existing work often stops at describing fragmentation. It fails to explore how coordination deficits causally affect downstream outcomes such as implementation strength, operator compliance, and utilization. This study contributes by defining and testing a simple mechanism linking coordination arrangements to utilization-related performance.

The third debate focuses on the gap between formal policy design and real-world behavioral responses. Policies may stipulate terminal-based operations, designated routes, or standard pickup points, but users and operators often adapt to minimize time, uncertainty, and costs. Behavioral studies show that passengers may prioritize proximity and speed over formal service nodes, while operators may avoid terminals when compliance reduces revenue opportunities or increases wait times. This mismatch is often exacerbated when enforcement is weak or inconsistent, when incentives are misaligned, or when terminals offer limited connectivity to actual travel demand. As a result, policies “on paper” can differ significantly from practice, resulting in persistent informal mismatches. Previous research has identified behavioral mismatches but often treats them as stand-alone explanations, rather than linking them to institutional conditions and upstream implementation. This study addresses the unresolved question of how formal policy design interacts with enforcement credibility and operator incentives to shape utilization outcomes.

The literature suggests that terminal effectiveness cannot be inferred from infrastructure provision alone, and that governance outcomes are generated through the interaction of institutional coordination, implementation credibility, and behavioral incentives. However, the existing evidence remains fragmented across separate discussions of assets, institutions, and behavior. A key gap is the lack of an integrated explanation linking these elements in a coherent causal sequence. Building on this debate, this study proposes a simple analytical logic: institutional coordination, power, implementation, participation/compliance of terminal utilization operators, and governance effectiveness. It uses qualitative evidence (interviews, observations, and documents) to demonstrate how upstream disruptions result in suboptimal utilization downstream and weakened governance outcomes.

To sharpen its disciplinary foundation, this review synthesizes literature at the intersection of transport governance and public administration/implementation studies, focusing on how institutional arrangements shape the real-world utilization of transport infrastructure. Three interrelated debates structure this review. First, research on the infrastructure-utilization gap highlights that capacity expansion does not automatically translate into increased ridership when access, service continuity, and user convenience remain misaligned. Second, research on institutional fragmentation and coordination failures demonstrates that overlapping mandates and weak intergovernmental routines erode implementation capacity, leading to inconsistent rules and weakened accountability. Third, research on the policy-behavior design mismatch explains why formal terminal operations can be bypassed when operator incentives, enforcement expectations, and everyday mobility practices favor informal pick-up points. Together, these debates motivate the paper’s primary explanatory focus: terminals underperform not because infrastructure is absent, but because the governance mechanisms that translate infrastructure into predictable and routinized service use are weak or contested.

To strengthen this paper’s scholarly dialogue with mainstream transportation research, this review also engages with the transportation systems and urban mobility tradition, which conceptualizes terminals as network nodes and evaluates performance through accessibility, connectivity, service levels, and system efficiency. Relevant studies in transportation planning and policy examine how terminal location, feeder network design, transfer penalties, service reliability, and competing informal pick-up points shape actual flows and user choices, often using indicators such as departures/frequency, passenger volume, route coverage, transfer times, and general travel costs. By integrating these transportation perspectives with institutional and implementation insights, this study positions suboptimal utilization not simply as an administrative issue but also as a problem of node performance and network integration mediated by governance, that is, coordination and enforcement influence whether operational designs produce observable system outcomes consistent with terminal function.

3 Methodology

This study employed a qualitative, descriptive approach. This approach systematically describes the reality of implementing the Anak Air terminal management policy, based on research findings. This allows researchers to broaden their understanding of the issue. The focus of this study is the effectiveness of Anak Air terminal management for optimizing public transportation and the inhibiting factors of this management in Padang City. Informants were selected using purposive sampling, a method of selecting informants based on their involvement and relevance to the implementation of the Anak Air Terminal management policy. This technique ensures that the data obtained is more accurate and aligned with the research focus. To obtain accurate and timely information, the researcher compiled a list of informants to be interviewed.

Qualitative interviews were conducted using a semi-structured interview design to capture the institutional and behavioral mechanisms that influence terminal governance and utilization. The interview guide was structured into four thematic blocks aligned with the study's analytical framework: (1) institutional governance and coordination, focusing on role clarity, inter-agency interactions, and decision-making routines; (2) enforcement and implementation practices, including oversight, sanctions, and operational consistency; (3) operator behavior and incentive structures, addressing compliance decisions, perceived costs and benefits, and informal practices; and (4) institutional and service context, encompassing infrastructure conditions, passenger demand, and perceived terminal legitimacy. This structure ensured comparability across respondents while allowing flexibility to explore context-specific explanations and causal narratives. Interviews were conducted with key stakeholder groups identified in Table 2, including transportation agency officials, terminal managers, law enforcement personnel, public transportation operators, and user representatives. Each interview lasted approximately 45–90 minutes, with an average duration of approximately 60 minutes, and was conducted in Indonesian at locations convenient to the respondents, including government offices and the Anak Air Terminal site.

Table 2. Research informants

No.	Informant Category
1	Land Transportation Management Agency of West Sumatra
2	Head of Terminal Management
3	Terminal Field Officer
4	Users of Terminal Services and Public Transportation
5	Public Transport Operators
6	Transportation Policy Expert
7	Trans Padang Coordinator

All interviews were audio-recorded with participants' consent and supplemented with detailed field notes. When recording was not permitted, contemporaneous notes were taken and expanded upon immediately after the interviews. To strengthen the validity of the data, interviews were supplemented with non-participant observation of terminal operations (vehicle entry and exit behavior, law enforcement presence, passenger flow) and analysis of relevant regulatory documents, operational guidelines, and official reports. These multiple sources were used for triangulation to reduce reliance on a single report and to link reported practices to observed behavior and formal institutional arrangements.

The qualitative data were analyzed using a clear, step-by-step coding process. First, all materials were compiled into a single data set and read repeatedly to understand overall patterns. Second, we applied open coding, labeling key statements and events with brief descriptive codes. Third, we grouped similar codes into broader categories through axial coding. For example, codes related to overlapping responsibilities and weak interagency communication were grouped as "weak coordination". In contrast, codes related to inconsistent monitoring and discretionary oversight were grouped as weak implementation/enforcement. Finally, the study used thematic synthesis to refine these categories into a small number of themes that were used to structure the Results and Discussion sections. Throughout the analysis, we used note-taking and triangulation across interviews, observations, and documents to ensure that interpretations were consistent and evidence-based.

To ensure transparency in our analysis, we used explicit criteria to translate raw qualitative data into analytical claims. Claims such as weak coordination were only made when similar evidence emerged repeatedly across different types of respondents and were supported by observations or documents. Similarly, claims of inadequate incentives were made when operators consistently described the net costs of terminal use without clear benefits. In short, the key claims in this paper reflect recurring patterns supported by multiple sources, not a single opinion. This approach provides a clear link from the coded data to the explanations presented in the Results and Discussion.

4 Operational Framework and Empirical Indicators

While this manuscript discusses governance effectiveness conceptually, empirical assessment requires a simple operational bridge between theory and observable evidence. Therefore, a concise operational framework is needed that connects core dimensions of terminal governance with measurable indicators and appropriate data sources. The goal is not to reduce qualitative analysis to statistics, but to determine which empirical signals support claims about effectiveness and its drivers. In this study, governance effectiveness is assessed through four observable dimensions: utilization, law enforcement, coordination, and connectivity, each represented by practical indicators that can be triangulated through interviews, observations, and documentary materials.

This framework is used in two ways. First, it guides data collection by ensuring that interviews, observations, and documents contain comparable empirical references. Second, it guides analysis by linking qualitative themes to observable indicators and evidential markers. Where direct administrative counts are unavailable, indicators are approached using supporting evidence, for example, repeated observational records of vehicle entry/exit and consistent stakeholder reports on departure frequency, combined with documentary records of enforcement actions and coordination activities. Details can be seen in the following Table 3.

Table 3. Operational framework linking dimensions, indicators, and data sources

Dimension	Illustrative Indicators	Primary Data Sources
Utilization	Daily departures; passenger volume; operator share	Observation; interviews; terminal logs
Law enforcement/ Implementation	Violation frequency; sanctions; inspection consistency	Officer interviews; reports; observation
Institutional coordination	Joint meetings; Memorandum of Understandings (MoUs); Standard Operation Procedures (SOPs); role clarity	Meeting minutes; decrees; interviews
Connectivity	Number of feeder routes; schedule alignment	Route maps; planning documents; interviews

5 Results

To avoid treating the institutional, behavioral, and operational dimensions as separate themes, the findings are presented in the order of the proposed causal chain. First, there is evidence of institutional coordination, with a focus on clarity of roles across agencies and alignment of operational priorities. Second, the robustness of implementation is examined, including the consistency of terminal-related regulations, monitoring routines, and service arrangements. Third, there is an analysis of operator participation/compliance as a behavioral mechanism linking policy to practice, highlighting incentives, perceived costs, and expectations of enforcement. Fourth, observed patterns of terminal utilization include how operator behavior and service continuity translate into passenger flows and terminal utilization. Finally, these utilization results are interpreted as indicators of governance effectiveness, noting how contextual factors moderate key relationships across the chain.

5.1 Anak Air Terminal Governance to Optimize Public Transportation in Padang City

Terminals, as land transportation hubs, play a strategic role in regulating vehicle flow, serving the community's mobility needs, and integrating modes. Therefore, good governance will determine the extent to which the terminal can fulfill its primary function, not only as a physical facility but also as a public transportation service center. This terminal is expected to become a hub for transportation activities, connecting intercity and interprovincial transportation (Inter-City and Inter-Provincial Transportation), intercity and interprovincial transportation (Intra-Provincial City Transportation), and local transportation modes such as public transportation and Trans Padang. The governance of Anak Air Terminal can be analyzed from the following aspects.

5.1.1 Terminal facilities

Terminal facilities are a crucial factor in supporting comfort, safety, and attractiveness for transportation users. Based on observations, Anak Air Terminal has several basic facilities that are quite comprehensive for a Type A terminal (Table 4).

Research findings confirm that the primary obstacle is not the availability of facilities, but rather the limited number of service users, which results in reduced terminal activity. Well-built facilities ultimately become suboptimal because people prefer other transportation options they consider more practical and faster, such as online services or travel agencies that pick up passengers directly at the location. Nevertheless, the terminal still generates income from

rental facilities, such as shop units, MSME units, and other supporting facilities. Rental of these kiosks and shops is one source of terminal revenue, in addition to vehicle fees. However, income from these facilities is relatively limited, as the number of tenants is not proportional to available kiosk capacity, and low passenger flow reduces the kiosks' attractiveness. Under these conditions, income from rental facilities for kiosks and shops is insufficient to cover the terminal's overall operational and maintenance costs. This indicates that facility utilization optimization still needs improvement through better governance strategies, such as partnership-based business management, promoting the terminal area as a center of economic activity, and integrating transportation services to increase passenger numbers.

Table 4. Facilities and infrastructure at Anak Air Terminal

No.	Supporting Facilities	Existing Conditions
1	Departure route	Available, but still lacking signs
2	Arrival route	Available and functioning
3	Passenger boarding and alighting area	Available and working
4	Parking area	Available and enough wide enough
5	Terminal office	Available and working with good
6	Counter sale tickets	Available (not yet active)
7	Customer service	Available
8	Disabled lane	Available
9	Toilet	Available
10	Canteen	Available (only 1 that is active)
11	China Central Television (CCTV)	Available
12	Tool, fire extinguisher, fire light (portable fire extinguisher)	Available

Further research findings clearly illustrate that although the physical facilities at Anak Air Terminal are considered adequate, their utilization rate is very low. Waiting rooms that should be filled with prospective passengers are often left empty, used only by staff or passengers in transit. Moreover, terminal activity is highly dependent on schoolchildren's mobility, so during school holidays or weekends, the terminal appears even more deserted. This phenomenon demonstrates that adequate facilities have not been a primary attraction for people to use public transportation at the terminal. This confirms that the main problem with Anak Air Terminal governance lies not only in the provision of physical facilities, but also in the involvement of the actors who are supposed to use those facilities. In other words, the availability of adequate facilities does not automatically guarantee terminal effectiveness unless consistent and sustainable use by users and transportation operators accompany it.

5.1.2 Human resources and operational management aspects

HR plays a crucial role in the successful management of a terminal. The presence of complete physical facilities will be meaningless without proper management by a competent workforce. Based on field observations at the Anak Air Terminal, the number of assigned staff remains relatively limited given the terminal's size and the complexity of the services it should provide. The staff's primary duties include administrative services, bus operational supervision, security, and maintaining the cleanliness of the terminal area. However, the workload is not evenly distributed because many terminal facilities are not yet operating optimally, leading some staff to be less active. The following Table 5 illustrates this information.

Based on intercity and regional transportation (Inter-City and Inter-Provincial Transportation) production data at Anak Air Terminal for January, a total of 176 buses arrived, carrying 2,036 passengers. Departures also showed a similar figure: 176 buses carrying 2,086 passengers and no transit activity. Based on this data, intercity and regional transportation (Inter-City and Inter-Provincial Transportation) traffic entering Anak Air Terminal remains relatively low. With an average arrival of only around 6 buses per day, this number is relatively small compared to the terminal's capacity and functions as an intercity and interprovincial transportation hub. This indicates that Anak Air Terminal is not yet functioning optimally in attracting intercity and regional transportation vehicles.

A comparison of capacity and baseline utilization further highlights the extent of suboptimal utilization. As a Type A terminal, Anak Air is designed to serve as a high-capacity intercity and interprovincial mobility hub, supported by extensive facilities and a wide range of service functions (interprovincial buses, interprovincial services, urban transportation, and Bus Rapid Transit (BRT) connectivity). In practice, observed traffic is well below what this functional capacity implies. Using operational numbers as a simple proxy for utilization, the annual figures for 2024 (1,919–1,921 movements) yield only about 5–6 bus movements per day on average, which is consistent with the daily pattern in January 2024. Furthermore, many registered operators operate only one daily departure, and some route segments remain inactive at the terminal, indicating that the terminal's physical and functional capacity (a multi-service hub) does not match actual operator participation and passenger flow. This gap between “designed function” and “realized use” reinforces the empirical claim that infrastructure provision alone does not

yield effectiveness without credible enforcement, aligned incentives for operators, and reliable feeder connectivity that makes terminal-based travel convenient for users.

Table 5. Total intercity and regional transportation data for January 2024

No.	Come	Passengers	Leave	Passengers
1	7	92	7	98
2	7	107	7	107
3	6	78	6	78
4	9	102	9	102
5	10	137	10	137
6	6	66	6	66
7	7	123	7	123
8	8	131	8	131
9	5	64	5	64
10	6	69	6	69
11	4	33	4	33
12	5	48	5	51
13	8	113	8	116
14	3	35	3	35
15	4	49	4	50
16	4	41	4	43
17	4	55	4	55
18	3	57	3	59
19	8	62	8	65
20	10	120	10	125
21	3	37	3	38
22	6	27	6	27
23	3	12	3	13
24	4	41	4	43
25	10	129	10	134
26	3	23	3	26
27	4	47	4	51
28	5	41	5	46
29	6	36	6	38
30	4	35	4	37
31	4	26	4	26
	176	2,036	176	2,086

Table 6. Total inter-city and inter-provincial transportation production data

No.	Year	Derpature	Arrive
1	2022	2,121	2,121
2	2023	773	982
3	2024	1,919	1,921

Based on Table 6, the trend analysis (2022–2024) shows highly volatile utilization rates, rather than a steady growth pattern. Annual numbers show a sharp contraction from 2022 to 2023, followed by a partial recovery in 2024. Specifically, arrivals decreased from 2,121 (2022) to 982 (2023) (-53.7%), while departures decreased from 2,121 (2022) to 773 (2023) (-63.6%). In 2024, utilization increased again to 1,921 arrivals and 1,919 departures, representing increases of 95.6% (arrivals) and 148.3% (departures) compared to 2023. However, the 2024 levels are still slightly below the 2022 baseline (approximately -9–10%), indicating that terminal utilization has not stabilized and remains sensitive to implementation dynamics, operator participation, and connectivity constraints. This volatility is analytically crucial because it suggests that terminal performance is less influenced by physical readiness and more by governance conditions that determine whether operators consistently come in and whether passengers reliably use the terminal.

Based on field observations, researchers found that Trans Padang is indeed the only transportation activity

still maintaining the terminal's function. However, Trans Padang passenger occupancy rates are relatively low, especially on holidays. User activity tends to increase only during school days, as most passengers are students transiting through the terminal. From the overall results of the interviews with informants, it was found that the operational implementation of the Anak Air Terminal is not yet running effectively and efficiently, even though the terminal has been physically equipped with facilities and infrastructure in accordance with the standards of a type A terminal. The levels of vehicle arrivals and departures remain low and fluctuate from year to year, indicating that terminal utilization is inconsistent and has not reached ideal occupancy. The minimal daily activity and low vehicle volume indicate that community involvement and cooperation with transportation operators are not optimal. This is reinforced by interviews with terminal managers and operators, who showed that most Intra-Provincial City Transportation and Inter-City and Inter-Provincial Transportation operators are reluctant to enter the terminal due to inadequate accessibility and the low number of passengers. In addition, only Trans Padang consistently operates at the terminal, even then with a relatively low passenger occupancy rate. Activity tends to increase only on school days, as most passengers are students or university students. Nevertheless, in terms of public service quality, the Anak Air Terminal has demonstrated excellent performance. The results of the Public Satisfaction Survey show that all service indicators, such as information, response, and infrastructure, received the predicate "very good". This indicates that the primary weakness lies not in services or facilities, but rather in effective implementation and low public participation.

Thus, the management of the Anak Air Terminal faces operational and social challenges. The lack of a comprehensive strategy, minimal route monitoring, and low public involvement have resulted in the terminal's facilities not being fully utilized. Ensuring the terminal's proper functioning requires active collaboration between stakeholders, improved accessibility, and a more effective communication strategy to build public awareness and participation.

5.1.3 Security and order aspect

Security and order are crucial elements in determining the success of a system or policy. Security serves not only to correct errors but also to ensure that implementation is in accordance with established objectives and standards. In the context of terminal management, supervision includes ensuring operator compliance with official routes, the proper use of terminal facilities, and the order and safety of public transportation services. Research findings indicate that despite technical supervision and outreach efforts, many operators still fail to comply with regulations, particularly among intercity bus (Intra-Provincial City Transportation) drivers. Common problems identified include: first, vehicles that are not roadworthy (especially in technical aspects such as brakes). Second, administrative omissions, such as incomplete inspection cards and route permits. Third, drivers' lack of awareness of the importance of safety procedures and operational legality. Data from ramp checks at Anak Air Terminal Type A from 2025 to July also support this statement. Of 1,106 arriving vehicles and 1,102 departing vehicles, the following were found: first, 74 technical violations (such as brake and light violations). Second, 177 inspection card violations. Third, 276 warnings issued, but no tickets or travel delays, indicating that supervision has not been accompanied by enforcement of concrete sanctions.

However, in practice at the Anak Air Terminal, supervision still relies on limited initiatives from the Land Transportation Management Agency of West Sumatra, without active involvement from the Padang City Transportation Agency, a regional partner, or from the Civil Service Police Unit and the National Police for field control. The research results show that Anak Air Terminal has successfully established a clean, safe, and comfortable public service image, particularly within the terminal. No indications of extortion, discrimination, or other irregularities were found based on user perceptions. However, management emphasized that on-board security remains the responsibility of each bus operator or driver. Therefore, while the terminal's services are optimal, further coordination with operators is needed to ensure comprehensive safety standards throughout the fleet.

5.1.4 Supervision aspect

Supervision in the management of the Anak Air Type A Terminal is still not optimal. Although regulations require all public transportation to enter the terminal, implementation is weak due to a lack of sanctions and low operator compliance. The terminal and the Land Transportation Management Agency of West Sumatra have conducted public awareness campaigns. Still, many operators, particularly the Intra-Provincial City Transportation, continue to fail to comply due to weak oversight and law enforcement. Ramp check results indicate numerous technical and administrative violations that are not followed up with firm action. Furthermore, terminal supervision has not actively involved regional agencies such as the City Transportation Agency, Civil Service Police Unit, or the National Police, leaving management entirely dependent on the Land Transportation Management Agency of West Sumatra, which has limited resources.

However, in terms of public services, the Anak Air Terminal has created a safe and comfortable environment. Surveys and interviews revealed no illegal levies or discrimination, and facilities such as restrooms, waiting rooms, and parking areas were deemed adequate. However, responsibility for passenger comfort and safety within the

fleet still rests with each operator. Therefore, although internal supervision has been implemented, its effectiveness still needs to be improved through: First, consistent enforcement of sanctions. Second, cross-agency coordination. Third, increasing operator compliance. Fourth, a more comprehensive monitoring approach from the terminal to the vehicle.

5.1.5 Inter-agency coordination aspect

Clarity in goal achievement strategies refers to concrete, systematic steps taken to achieve organizational or program objectives optimally, effectively, and efficiently within a relatively short timeframe, while ensuring predetermined goals are met. Anak Air Terminal should have a clear strategy to address the challenge of minimal facility utilization by the public and public transportation operators. Therefore, the Land Transportation Management Agency of West Sumatra, or terminal management, must make concrete efforts to formulate concrete strategic stages and approaches to optimally reactivate the terminal's functions. This strategy needs to be collaborative, well-planned, and tailored to the social and geographical conditions of Padang City.

To optimize the management of Anak Air Terminal, a strategy must be designed and implemented sustainably. Researchers assess that these steps represent a creative, unconventional strategy aimed at changing public perception of the terminal. However, the effectiveness of this strategy remains limited because personal branding or the terminal itself has not yet reached the general public. Research findings indicate that, despite the implementation of a socialization strategy, technical and infrastructure factors remain major obstacles to its continued success. Efforts to revive the terminal are not enough to rely solely on inviting bus operators (POs); they must also be balanced with physical improvements and targeted incentive policies. As shown in the image above, if public awareness campaigns are conducted to direct drivers directly to the terminal, it will also be ineffective. Why is this? When buses enter the terminal, if the passengers aren't there, it's the same. Therefore, officers must also ensure that passengers are using the service.

The research findings explain that the government's incentive strategy remains insufficiently attractive to operators, especially when concrete solutions don't accompany it on the ground. As a solution, providing free Damri buses as a connecting (feeder) mode underscores the importance of active government intervention to bridge the connectivity gap that cannot be addressed solely through regulation or outreach. With a feeder directly connected to the city center, Anak Air Terminal can be more accessible to the public, making the terminal's existence relevant and attractive to both passengers and operators. This confirms that the strategy's success depends not only on administrative policies but also on tangible ease of access and functional incentives on the ground.

Interviews indicate that the main challenge lies in passenger preferences. Despite public awareness campaigns about entering the terminal, the reality on the ground shows that many passengers still prefer to wait on the roadside or at unofficial locations deemed more convenient. In fact, he said, it's actually the users themselves who create Illegal terminals, as they find these informal points easier to reach than going to Anak Air Terminal. This creates a dilemma for operators: entering the terminal means potential losses, as there are no passengers there. Therefore, they prefer to stay at the pool or alternative locations that are more effective at attracting passengers.

Based on interviews with various informants, it can be concluded that the strategy to achieve Anak Air Terminal's goals has not been effective. Although several efforts have been made, such as collaborating with Trans Padang, organizing social activities within the terminal area, and renting facilities as a form of outreach, these strategies remain partial and unstructured and fail to address the root of the problem: low terminal utilization by operators and the public. Complaints from drivers and bus companies regarding narrow road access, the absence of passengers at the terminal, and the public's tendency to use Illegal terminals indicate that there is no strategic approach capable of changing the behavior patterns of both users and transportation service providers. Furthermore, coordination among stakeholders, including the Land Transportation Management Agency of West Sumatra, local governments, transportation operators, and the community, remains suboptimal. Therefore, it can be concluded that the strategies implemented to manage the Anak Air Terminal have not been effective in supporting the achievement of its primary objective as a public transportation hub in Padang City. A more planned, collaborative, and needs-based strategy is needed, including the provision of feeder transportation, increased physical accessibility, and adaptation to the city's geographic and social conditions.

5.1.6 User satisfaction and perception aspects

The effectiveness of air terminal management in optimizing public transportation in Padang City can be seen from the extent to which management has achieved its goals and objectives. These goals will be achieved more smoothly, orderly, and effectively if management has instilled a deep awareness and confidence in achieving governance objectives. Research findings indicate that terminal management has implemented a public service evaluation mechanism through the Public Satisfaction Survey (PSS). Based on available PSS data, all service aspects received a rating of "Very Good", with index scores ranging from 96.25 to 97.25. Aspects such as Information, Response, Consultation, and Complaints received the highest scores, indicating that administrative services and user communication are considered satisfactory by the public. This reinforces the research finding that, from an internal

and administrative perspective, the quality of terminal services is already very good. However, the main challenge remains the low participation of operators and the public, rather than the quality of service itself. The success of public service provision is determined not only by the completeness of facilities and service standards, but also by the effectiveness of outreach and community involvement. The Anak Air Terminal has adequate infrastructure and services. Still, it has not become the public's primary choice due to low visibility, a lack of active operators, and the lack of an integrated strategy to attract users sustainably.

The main obstacle faced is preventing the Anak Air Terminal, which has been significantly overspent, from becoming a useless terminal like its predecessor. Furthermore, the absence of adequate connecting modes at the terminal, such as public transportation, which is rarely accessible to the terminal, means that the integration designed normatively in policy has not been realized in practice. Research findings indicate that regulatory support alone is insufficient without supervision, incentives, or strict sanctions for violating operators. This indicates weak implementation in terminal governance. These findings indicate that Trans Padang's role as a connecting mode is not optimal due to a lack of coordination between operators and limited resources.

The current condition of the terminal reveals a gap between normative objectives and on-the-ground implementation. Due to the nighttime bus schedule, the schedules between these modes are not synchronized, as Trans Padang operates until 6:00 p.m. In public transportation planning theory, intermodal integration is a fundamental principle in managing Type A terminals. However, the success of this objective is determined not only by policy documents but also by synergy between service providers, synchronized operating schedules, and the availability of truly accessible connecting modes. If only some parties fulfill their roles, the objective becomes merely symbolic. Although these objectives have been clearly defined in documents and statements by policy managers, implementation on the ground has not shown optimal achievement. One key indicator is the absence of intermodal bus operators entering the terminal, even though they are legally required to pick up and drop off passengers there. The involvement of intermodal bus modes is a key element in creating integrated services and intermodal connectivity. While Trans Padang operates at the terminal via corridors 1 and 4, the availability of other connecting modes, such as public transportation, remains inadequate, thus failing to support comprehensive integration. This clearly stated objective is neither optimal nor effective. This situation demonstrates a serious gap between the formulated normative objectives and actual implementation on the ground. While formal clarity of objectives exists, it is not accompanied by a comprehensive and synergistic implementation strategy across relevant parties. Therefore, the indicator "clarity of objectives to be achieved" does not reflect optimal governance effectiveness, as not all transportation actors are fulfilling their mandated roles, and not all user needs are being accommodated in an integrated manner. The outlined objectives are not yet aligned with user needs and behavior.

5.2 Factors Inhibiting the Effectiveness of Anak Air Terminal Governance for Optimizing Public Transportation in Padang City

Research findings indicate that four indicators are the main obstacles to optimizing the function of Anak Air Terminal.

5.2.1 Organization

Organizations managing terminals and public transportation must have a flexible yet coordinated structure and an efficient bureaucratic system to support operational functions and public services. In the context of terminals, this includes how the managing institution can respond to field dynamics, coordinate among work units, and implement regulations and procedures quickly and accurately. The managing organization for Anak Air Terminal is the Land Transportation Management Agency of West Sumatra, which falls under the Ministry of Transportation of the Republic of Indonesia. Although it has technical authority over the management of terminal facilities and infrastructure, the Land Transportation Management Agency of West Sumatra lacks direct authority over several other important aspects.

Based on the research findings, the main obstacle to optimizing the Anak Air Terminal's function lies in its accessibility. Access roads to the terminal remain narrow, densely populated, and unsuitable for large vehicles such as intercity buses. Furthermore, low-hanging power lines along the entrance road pose a technical obstacle that endangers vehicle safety. Plans for land acquisition and road widening have been included in the Padang City Government's medium-term agenda, but have yet to materialize because the process depends on decisions and budgeting from the Padang City Government. This is further complicated by the rising land prices around the terminal, which have now doubled or tripled, requiring a significant budget allocation.

Real challenges in terminal operations, particularly in HR and effective coordination among stakeholders, remain. He stated that currently, there are only two field officers tasked with directing intercity and intracity buses into the terminal. This very limited number of personnel undoubtedly complicates vehicle monitoring and enforcement in the field, particularly in dealing with operators who are reluctant to comply with official route regulations. Although the Land Transportation Management Agency of West Sumatra, including the Head of the Office and structural officials, has attempted to coordinate through meetings with representatives of West Sumatra Bus Companies to

provide important guidance on optimizing terminal utilization, the results have not been fully effective. The reality on the ground shows that only a handful of Inter-City and Inter-Provincial Transportation buses and the Trans Padang fleet regularly use the terminal. In contrast, Intra-Provincial City Transportation buses remain reluctant to enter, even though they are required to do so by regulation.

5.2.2 Environment

External conditions greatly influence the effectiveness of transportation management. The success of environmental-organizational relationships depends on three variables: the level of environmental predictability, the accuracy of environmental perceptions, and the level of organizational rationality. Research shows that management has attempted to establish cooperation and provide supporting facilities, but limited information reach and a lack of visibility have resulted in the terminal being poorly known. This has led to low public visits and suboptimal facility utilization. Based on information media, such as Instagram, not many people follow it, so the information does not cover the entire surrounding environment. Based on field findings, there are three main environmental challenges:

First, low environmental predictability. The public lacks sufficient information or is even unaware of the existence of the Anak Air Terminal. The lack of public awareness campaigns by management has led people to opt for more accessible modes of transportation, such as taking the bus from the roadside or the nearest bus stop, making the terminal a less common option.

Second, an inaccurate perception of environmental conditions. The government and terminal management appear not to fully understand the actual behavior of Padang City residents, who tend to choose practical and time-saving transportation options. As a result, the implemented strategies have not addressed the root of the problem, such as the public's habit of using illegal terminals, which they consider closer and more efficient.

Third, a low level of organizational rationality in responding to environmental dynamics. The organization's response to conditions on the ground has not been fully strategic. Although they have been put in place, such as the collaboration with Trans Padang and the provision of business space at the terminal, these policies have not been widely accessible to the public. On the other hand, long-distance bus operators also have rational reasons for avoiding the terminal, such as distance, high operational costs, and a lack of passengers on-site. Based on a series of interviews, it can be concluded that the problem of low utilization of the Anak Air Terminal lies not only in infrastructure but also in social and behavioral dimensions. The public is not accustomed to using the official terminal because of established transportation patterns. Furthermore, Intra-Provincial City Transportation operators are also reluctant to comply with terminal entry requirements due to low demand and administrative unpreparedness. Therefore, for the Anak Air Terminal to function optimally, the solution approach cannot simply consist of providing physical facilities. It must also be accompanied by strategies that address user behavior, increase transportation literacy, and strengthen collaboration between central and regional governments and operators.

Fourth, HR directly involved in public services play a crucial role in service effectiveness. In the context of terminal management, this includes the skills, competencies, motivation, and professionalism of the HR involved in operations and services. In the management of the Anak Air terminal, field workers, including terminal officers, vehicle guides, and administrative staff, play a central role in ensuring the terminal operates properly. However, findings in the field indicate that worker characteristics also act as a hindrance to effective terminal governance. Fifth, management characteristics encompass strategies, policies, and work mechanisms designed to efficiently manage resources to achieve organizational goals. In the terminal context, effective management must be able to formulate adaptive policies, establish cross-agency coordination, and perform continuous control and evaluation. At the Anak Air Terminal, suboptimal management practices are a hindrance, particularly in planning, operational implementation, inter-agency coordination, and responding to field dynamics.

6 Discussions

This discussion is structured around the mechanisms by which governance arrangements translate into utilization outcomes. Empirical patterns suggest that institutional coordination is a necessary upstream condition: when institutions lack a shared operational mandate, implementation becomes fragmented and selectively applied. This weak implementation strength diminishes the credibility of terminal-based rules and lowers the expected returns to compliance for operators. Consequently, participation by key behavioral link operators declines, leading to low terminal utilization even when certain internal service aspects are deemed adequate. In turn, low utilization undermines governance effectiveness by limiting service integration, weakening the legitimacy of terminals as key nodes in public transport operations, and limiting the system's capacity to deliver predictable, regular mobility outcomes. In more detail, this section interprets the findings through three complementary lenses: governance theory, transport systems theory, and institutional analysis, to go beyond descriptive accounts of low utilization. From a governance perspective, the case of Anak Air Terminal illustrates how formal institutional design and infrastructure investment do not automatically produce effective service outcomes when coordination and implementation mechanisms are weak. From a transport systems perspective, the terminal functions as a network node whose performance depends on reliable route integration, intermodal connectivity, and predictable operational

routines. From an institutional perspective, utilization is a matter of collective action and incentives: operators and users respond rationally to prevailing rules (enforcement credibility, time costs, revenue uncertainty) rather than formal rules on paper. These lenses help explain not only what is happening (low utilization) but also why an infrastructure-led approach fails to produce the desired governance outcomes in this context.

The purpose of establishing the Anak Air Terminal as a transportation hub and intermodal integration hub has been clearly formulated in the policy document. This aligns with Minister of Transportation Regulation No. 32 of 2015, which governs type A terminals as safe, orderly, and structured centers for integrated travel and public services. However, the implementation of this objective has not yet shown optimal results in practice. Several indicators support this conclusion: First, intercity bus operators (Intra-Provincial City Transportation) are not fulfilling their obligation to enter the terminal, even though they are required by regulation to do so for operational order. Second, Trans Padang, as a connecting mode, has not been able to fully fulfill its integrative role due to fleet limitations and an operational schedule that is not synchronized with the arrival of intercity buses, particularly at night. Third, intermediate transportation modes such as public transportation are not available at the terminal, making it difficult for passengers to reach their final destinations. Fourth, from the perspective of service users, there are still gaps in accessibility and information, making it difficult for passengers to find connecting transportation.

On the other hand, although the West Sumatra Land Transportation Management Agency and terminal management have conducted outreach and coordination between supporting agencies and institutions, such as the Padang City Government, the City Transportation Agency, and bus operators, it has not been optimal. As a result, the effectiveness indicators for the Anak Air Terminal governance objectives have not been fully achieved. These objectives have been well-formulated formally, but their implementation has not been effective because: first, there is no full involvement of transportation operators. Second, intermediate modes are not yet available in sufficient quantities and on schedule. Third, Inter-agency coordination is not yet synergistic. Fourth, the implementation strategy is not based on actual user behavior. Therefore, although the direction and vision for terminal development are in accordance with regulations, the gap between objective formulation and implementation is a major obstacle to optimizing public transportation in Padang City via the Anak Air Terminal.

The Anak Air Terminal's management strategy to encourage public and operator utilization of the terminal has not been implemented effectively or comprehensively. Strategic efforts undertaken, such as collaborating with Trans Padang, holding social activities within the terminal area, using social media, and renting terminal facilities, are innovative but partial and do not address the root of the problem. For example, these social activities do introduce the terminal to the public, but they are not accompanied by a long-term strategy to ensure changes in public transportation behavior [32–35]. The main obstacles identified in the field include: first, low visibility and physical accessibility to the terminal, due to narrow entrances and inadequate supporting infrastructure. Second, disinterest among intercity bus operators in entering the terminal, as they perceive no passengers waiting inside, and that pool or roadside locations are more economically strategic. Third, the absence of real government intervention in the form of functional incentives, such as providing feeder modes or fuel support, to encourage operators to continue using the terminal even when it is not full. Fourth, inadequate socialization and branding of the terminal, especially on digital platforms and local media, result in many people still being unaware of its existence.

Operators, such as Intra-Provincial City Transportation, stated that public behavior itself drives the creation of Illegal terminals, as passengers prefer to wait on the roadside or at informal points. Strategies that rely solely on calls to enter the terminal, without restructuring routes and empowering official pick-up points, tend to be unsuccessful. Therefore, the current strategy remains top-down, unresponsive to social conditions, and has not yet created a transportation ecosystem that supports comprehensive intermodal integration. There is no real synergy between user needs, the role of operators, and government policy direction [36–38].

6.1 Policy

This indicator assesses the extent to which the organization's policies are based on comprehensive, participatory analysis. Based on research findings, the development of the Anak Air Terminal emphasized the physical aspects without considering its operational systems and connectivity. There is no roadmap or structured plan for how transportation integration will be developed after the terminal is completed. Land Transportation Management Agency of West Sumatra, as the operator, also faces limited coordination with the Padang City Government, particularly regarding road access and regulating Illegal terminals. Furthermore, policies concerning forcing bus operators to enter the terminal are not accompanied by strict sanctions mechanisms or attractive incentives. The policy formulation process, which is not based on a study of user behavior and operator interests, is the reason why existing policies are ineffective. This indicates that governance effectiveness will be very low if the analysis and policy formulation process does not address the root of the problem.

6.2 Planning

Thorough planning encompasses short-, medium-, and long-term planning that takes into account geographic, social, and available resource conditions. In the context of the Anak Air Terminal, terminal construction planning is underway, but accessibility aspects have not been fully addressed. Narrow access roads are a major obstacle that remains unresolved, as responsibility for this falls to the Padang City Government. In terms of operational management, planning also appears immature. The number of management personnel is limited, active routes are very few, and there is no specific plan to attract Intra-Provincial City Transportation operators back to utilize the terminal facilities. Furthermore, there is no clear plan for the integration of additional transportation modes that can support passenger connectivity to and from the terminal. This indicates that the planning process focuses solely on the physical aspect (terminal building) without considering operational management, community social aspects, and cross-sectoral collaboration [39–42]. As a result, the terminal remains a facility that is not optimally utilized for its intended purpose.

6.3 Facilities and Infrastructure

Facilities and infrastructure are a crucial foundation for supporting the effective management of an organization or public service, including terminals. Based on field findings and interviews with various informants, Anak Air Terminal physically meets the standards of a Type A terminal, with adequate facilities such as waiting rooms, restrooms, parking areas, disabled access, CCTV, and ticket counters and kiosks. However, the availability of these facilities is not commensurate with their utilization rate. Many facilities are underutilized, such as empty kiosks, a single active canteen, and ticket counters that are not yet fully operational. Even spaces such as offices, customer service, and departure lanes often appear empty of passenger activity. This situation indicates that the presence of infrastructure without robust operational activity will only leave the terminal as an empty physical space lacking functional value. On the other hand, external infrastructure, such as road access to the terminal, poses a major obstacle that significantly impacts operational performance [43–45]. Narrow access roads, dense residential areas, and low cables that pose a danger to large intercity buses are key inhibiting factors. Some operators prefer private pools near the city center, perceived as safer, more strategic, and attracting more passengers. This situation hinders the goal of integrating transportation services through the Anak Air Terminal. In addition to technical obstacles, the terminal's location on the outskirts of the city and its lack of connections to other modes of transportation (such as feeder buses or public transportation) also require people to make multiple trips to reach the terminal. This is considered inefficient and discourages people from directly utilizing the terminal's facilities. The research findings reinforce the finding that the primary problem lies not in the availability of internal facilities, but rather in weak external accessibility and the lack of economic or operational activities that support the terminal's dynamic operation. Even additional economic potential, such as kiosks and canteens, remains untapped due to the lack of significant passenger traffic.

6.4 Implementation

Implementation is a crucial stage in the governance cycle, as it is where policies, objectives, and strategies are tested in practice. Based on field findings, the management of the Anak Air Terminal has not been effective and efficient, even though infrastructure and service standards have been prepared in accordance with regulations for Type A terminals. Data on vehicle arrivals and departures show significant fluctuations from year to year, with the lowest figures occurring in 2023. This indicates that the terminal's occupancy and activity levels are still far from ideal. One of the main causes is the low participation of intercity and intercity transportation operators, who tend to pick up passengers at the pick-up point or roadside rather than entering the terminal. The underlying reasons for this are not solely technical, but also due to the absence of passengers inside the terminal and its perceived lack of strategic accessibility.

Conversely, although Trans Padang is the only mode of transportation operating regularly at the terminal, passenger occupancy rates are low, particularly outside of weekdays and school holidays. This indicates that the intermodal hub function has not been realized, as there is no continuity between the arrival schedules of intercity and intercity transportation and the connecting modes, and there is no feeder strategy bridging the terminal with the city center. However, in terms of internal public services, the Anak Air Terminal has demonstrated excellent performance. The results of the Public Satisfaction Survey showed an index value above 96.25, reflecting highly satisfactory service quality in terms of information, responsiveness, and facility management. Interviews with service users also indicated satisfaction with the comfort, cleanliness, and completeness of terminal facilities. This means that the main problem with implementation is not the technical side of the service, but rather the minimal utilization of facilities by the public and operators. Statements from operators who prefer to wait outside the terminal reinforce the conclusion that low terminal activity is not due to management's unpreparedness, but rather because the public has not yet made the terminal a primary point of movement. This is also due to a lack of socialization, a weak communication strategy, and the absence of a consistent incentive approach or policy enforcement [37, 46–48].

6.5 Supervision

Supervision is a crucial aspect in ensuring effective terminal governance, not only in terms of regulatory compliance but also in ensuring the safety, comfort, and regularity of public transportation services [49–51]. However, based on field findings, supervision at the Anak Air Terminal is still not optimal, both systematically and operationally. Although regulations stipulate that all public transportation, both long-distance and short-distance, must enter the terminal, implementation remains weak due to a lack of strict sanctions, limited personnel, and low levels of operator compliance, particularly among long-distance operators. As reported by the Land Transportation Management Agency of West Sumatra and field officers, outreach has been conducted, and direct warnings have even been issued in the field. However, because it is not accompanied by firm law enforcement instruments, these efforts have not been able to comprehensively change operator behavior. Furthermore, the results of the 2025 ramp check also revealed a significant number of violations (both technical violations, such as brake and light violations, and administrative violations, such as inspection cards). However, all violations were met with only warnings, without legal action such as tickets or departure delays. This reinforces the finding that field supervision has not been accompanied by the courage to take action. However, in terms of monitoring public services within the terminal, such as cleanliness, security, and comfort, Anak Air Terminal is considered to have carried out its oversight function well. Based on user interviews and surveys, no indications of extortion, discrimination, or service irregularities were found. Facilities such as waiting rooms, restrooms, and parking areas are also well-maintained. However, it should be noted that monitoring comfort and safety within the bus fleet remains the responsibility of each operator and has not been actively monitored by terminal authorities. This means that service standards have not yet fully addressed the entire passenger journey, especially after boarding.

6.6 Factors Inhibiting the Effectiveness of Anak Air Terminal Governance for Optimizing Public Transportation in Padang City

The first factor is organization, which relates to organizational structure, coordination flow, bureaucratic system, and managerial flexibility. The Anak Air Terminal management organization is the Land Transportation Management Agency of West Sumatra, which is part of the Ministry of Transportation. Based on research results, it was discovered that the management structure tends to be vertical and separate from the authority of the Padang City Government, particularly regarding road access, land acquisition, and traffic management. The lack of full authority prevents terminal management from directly intervening in many critical aspects that ultimately determine the terminal's functionality. Coordination between the Land Transportation Management Agency of West Sumatra and the Padang City Transportation Agency and transportation operators also remains weak. There is no regular communication forum and no formal agreement between agencies to support the smooth operation of the terminal. This has resulted in various issues, such as the lack of involvement of Intra-Provincial City Transportation operators, the lack of feeder modes, and unclear oversight of Illegal terminals that have never been fully resolved. Thus, an overly centralized organizational structure, minimal horizontal coordination, and inflexibility are major obstacles to the effective management of the Anak Air Terminal.

The second factor is the social, cultural, and geographical environment, which significantly influences the behavior of transportation users. Padang residents have a culture of practicality and high efficiency in their daily activities, including mobility. They prefer to ride from the roadside or nearby locations rather than travel to distant, difficult-to-reach terminals. From the perspective of long-distance bus operators, illegal terminals are considered more advantageous because they save fuel and time and are closer to passenger gathering points. This is a response to environmental conditions that do not support the existence of official terminals. Therefore, the unfavorable social and geographic characteristics of the environment, along with the weak government response to these conditions, are significant external factors that hamper the effectiveness of terminal governance.

The third factor is the HR directly involved in terminal operations, who play a crucial role in ensuring optimal service. However, at the Anak Air Terminal, the number of personnel on duty is very limited. Only two officers handle all daily operations, including supervision and service, as well as facility management. The lack of personnel makes it impossible to actively monitor activities outside the terminal, such as buses operating at illegal terminals. Consequently, enforcement efforts and outreach efforts to drivers and POs are ineffective and rely solely on informal appeals. Furthermore, there is no training or capacity-building for terminal staff in operational management, public services, or the use of information technology. This makes management appear passive and less adaptable to emerging challenges in the field. The limited number of employees, capacity, and system support are significant internal factors hampering effective terminal management.

The fourth factor is management, encompassing work strategies, control systems, and internal and external coordination mechanisms. At Anak Air Terminal, the implemented strategy fails to address the root of the problem. For example, terminal promotion involves renting space for public events. Still, it is not accompanied by a systemic approach, such as route integration, mode-scheduling adjustments, or operational subsidies for feeder modes. The lack of an integrated management system, whether through digital technology or stakeholder coordination, results in

a slow, reactive decision-making process. There is no regular evaluation system used to improve performance, and no performance indicators to measure the terminal's operational success. Oversight functions are also not implemented comprehensively, including internal oversight of staff and facilities, as well as external oversight of transportation operators. This has led to regulatory violations, such as the use of Illegal terminals or drivers lacking administrative documents, being ignored. In general, the management of Anak Air Terminal does not reflect the principles of effectiveness, efficiency, and accountability that should underpin the management of public facilities [52–55].

6.7 Institutions Aspect

Research results indicate that institutional governance at Anak Air Terminal continues to face various challenges, particularly in inter-agency coordination. This terminal falls under the authority of the Ministry of Transportation through the Land Transportation Management Agency of West Sumatra. At the same time, the local government, through the Padang City Transportation Agency, also has an interest in regulating transportation within its territory. This duality of authority leads to overlapping roles and ineffective terminal operational policies. In the field, terminal managers admitted to having conducted outreach to intercity and intercity bus operators (Inter-City and Inter-Provincial Transportation) and to intra-provincial city bus operators (Intra-Provincial City Transportation) to use official terminals. However, most operators still choose to use illegal terminals due to accessibility and economic considerations. This indicates weak institutional capacity for regulation and oversight. Thus, the institutional governance of the Anak Air Terminal has not been implemented in accordance with good governance principles. To improve its effectiveness, strengthened coordination between the Land Transportation Management Agency of West Sumatra and the Padang City Transportation Agency (Astra Transportation Agency), as well as the active involvement of transportation operators and service users, is needed. Without strong institutional support, the terminal will not function optimally as a public transportation hub.

6.8 Public Services Aspect

The services and operations at Anak Air Terminal are generally well-designed. MSME facilities, ticket counters, parking areas, restrooms, prayer rooms, and waiting rooms are available to support passenger comfort. Furthermore, the terminal is equipped with surveillance systems, including CCTV, and a rest area for the vehicle crew. However, based on interviews and observations, utilization of these facilities remains very low. On average, only three to five buses enter the terminal per day, and the number of passengers is also limited. This situation indicates that service and operational governance are not yet optimal. The public transportation services must meet the basic principles of regularity, punctuality, safety, comfort, and affordability [56–58]. At Anak Air Terminal, regular schedules and vehicle flow have not been achieved because most operators choose to pick up and drop off passengers outside the terminal. This has resulted in low utilization of terminal facilities and undermined public trust in the terminal's function.

Furthermore, the theory of public service standards emphasizes that service quality can be measured through aspects of reliability, responsiveness, assurance, empathy, and physical facilities [59–61]. Field findings indicate that the terminal's physical facilities are adequate, but reliability and responsiveness remain elusive. The lack of a digital information system regarding bus schedules, a shortage of service personnel, and limited access to the terminal worsen operational issues. Consequently, the service and operational governance of Anak Air Terminal is ineffective. To optimize service, operational improvement strategies are needed, including integrating routes with local transportation, providing technology-based service information, and strengthening oversight of operators to ensure they are registered as official terminals. Without these measures, terminal services will not attract the public and transportation operators.

6.9 Financial Aspect

The financial aspect is a crucial indicator of terminal governance, as it concerns how funds are obtained and managed to support daily operations, facility maintenance, and the sustainability of public services [62–64]. Research shows that Anak Air Terminal generates primary revenue from two sources: vehicle fees and rental of facilities such as MSME units, business space, and shops. However, in 2025, vehicle fees will be eliminated to attract intercity and interprovincial buses (Inter-City and Inter-Provincial Transportation) and intra-provincial city buses (Intra-Provincial City Transportation). Therefore, revenue from this sector remains relatively small. The data obtained indicates that the occupancy rate of MSME kiosks is still low, with some units even empty due to low terminal visitor numbers. This situation has reduced the attractiveness of the terminal area to prospective businesses. Furthermore, vehicle fees are minimal because few intercity buses actually use the official terminal as a departure or arrival point.

This finding aligns with research [65] on terminal fees in Majalengka, which states that weak oversight, low operator compliance, and the presence of Illegal terminals contribute to unmet revenue targets. The good public financial governance should ensure accountability, transparency, and fiscal sustainability in the management of public facilities [66–68]. At the Anak Air Terminal, transparency is evident in official financial reports, but accountability

remains elusive because revenue falls well short of its potential. With minimal revenue, the terminal struggles to fund its operations independently. Consequently, facility maintenance relies heavily on the Ministry of Transportation's budget. This shows that the financial management of Anak Air Terminal is not yet effective and has not yet become an independent source of income, both for terminal management and for contributing to Regional Original Income. Therefore, strategies to improve financial governance need to be directed at: (1) optimizing the use of MSME units by collaborating with local business actors, (2) regulating Illegal terminals so that vehicle levies are maximized, and (3) innovation in revenue sources, for example the use of terminal space for social, commercial activities, or rental of advertising space.

6.10 Facilities and Infrastructure Aspect

Observations indicate that the Anak Air Terminal has adequate facilities and infrastructure to support public transportation. These facilities include a terminal building, passenger waiting room, ticket counter, prayer room, restrooms, parking area, MSME units, and supporting facilities such as a lactation room and a rest area for vehicle crews. Furthermore, the terminal is equipped with CCTV surveillance and a security post. Although the facilities are adequate, their utilization rate remains very low. This is due to the limited number of buses entering the terminal, resulting in underutilization of existing facilities. Some MSME units are empty due to low passenger traffic, and the waiting room is rarely used, as passengers prefer to board or disembark at the designated terminal or directly from the bus agent. This situation indicates underutilization of the terminal's facilities and infrastructure. In addition to internal issues, supporting infrastructure around the terminal also poses a challenge. The road to the Anak Air Terminal is relatively narrow and not strategically located, making access for large vehicles difficult. This situation has led bus operators to prefer using the designated terminal or alternative locations considered more accessible to passengers. Furthermore, technical issues such as inadequate electrical cables and telecommunications networks remain, potentially disrupting the safety and aesthetics of the terminal area. The transportation infrastructure will not provide optimal benefits without integration with operational systems and consistent policy support [69–71]. This aligns with the perspective, which emphasizes that public facilities must be managed in accordance with principles of effectiveness and efficiency to truly support service objectives [72]. At the Anak Air Terminal, existing facilities have not yet provided added value to the community due to suboptimal service management and operational regulations.

This situation demonstrates that the management of the Anak Air Terminal's facilities and infrastructure remains limited to physical development, without a sustainable utilization strategy. To address this, steps are needed, such as: (1) improving road accessibility to the terminal, (2) reorganizing the cable network and utilities in the terminal area, (3) promoting the terminal as a transportation service center, and (4) routine maintenance to ensure facilities remain ready for use when passenger traffic increases. Thus, the facilities and infrastructure of the Anak Air Terminal can be said to be "sufficiently available" in terms of quantity, but "not yet effective" in terms of governance, as they do not support the expected optimization of public transportation.

6.11 Security Aspect

Security is a crucial aspect of terminal management, as it directly relates to passenger comfort, traffic order, and smooth vehicle operations. At Anak Air Terminal, security measures include security posts, CCTV installations at several strategic locations, and officer deployments to monitor activities within the terminal area. However, observations and interviews indicate that these security functions are not yet functioning optimally. One major obstacle is the low level of activity at the terminal, which often leaves security officers without a clear focus. Furthermore, the widespread use of Illegal terminals outside Anak Air Terminal weakens oversight of passenger safety and security, as these activities occur outside the reach of terminal authorities. The effectiveness of a security system depends not only on the availability of physical facilities but also on the institutional capacity to regulate and enforce regulations [73–75]. At Anak Air Terminal, weak coordination among terminal management, the police, and the Padang City Transportation Agency led to the security system not functioning as intended.

Furthermore, transportation security standards set by the Ministry of Transportation include regular vehicle ramp checks, monitoring passenger activity, and clearing the terminal area of informal activities that could disrupt comfort. Field evidence indicates that some of these activities have not been implemented consistently, resulting in low public perception of terminal security. Consequently, security governance at Anak Air Terminal still has weaknesses in implementation and coordination. To improve security, strategies are needed, including: (1) strengthening cooperation between terminal management, the Transportation Agency, and the police; (2) increasing and equalizing the number of security personnel in the terminal area; (3) stricter enforcement of regulations against operators and the public using illegal terminals; and (4) more effective use of surveillance technology. If these measures can be implemented, the security of Anak Air Terminal will be more assured, thereby increasing public trust in using public transportation through official terminals.

6.12 The Effectiveness of Terminal Governance: Theoretical and Empirical Perspectives

The findings show that the effectiveness of the Anak Air Terminal's governance remains low, even though the terminal's physical infrastructure meets the standards of a Type A terminal. The effectiveness is determined by how far an organization can achieve its goals through efficient resource use and coordinated actions [19, 76, 77]. In this context, the primary goal of the Anak Air Terminal is to serve as a hub for multimodal integration and to enhance community mobility. However, empirical evidence indicates a significant gap between normative policy goals and factual implementation outcomes.

Conceptually, terminal effectiveness should not be assessed merely from physical or infrastructural readiness but from how far it fulfills its role as a node of social and economic mobility [78–80]. A terminal is not merely a place for loading and unloading passengers but also a “node of connectivity” that stimulates local economic, social, and transport interactions. The low number of operating buses and passengers indicates that the Anak Air Terminal has not yet achieved this role. This aligns with the output–goal relationship concept: effectiveness is weak when the tangible outputs fail to align with the intended organizational goals [81, 82].

6.13 The Gap between Infrastructure and Utilization

Empirical data show that Anak Air Terminal possesses adequate facilities: waiting rooms, parking areas, restrooms, disabled access, CCTV, and MSME stalls. However, the actual utilization of these facilities remains far below capacity. The public sector effectiveness theory suggests that the success of public infrastructure projects lies not in the physical presence of facilities, but in the institution's capacity to generate social utility [73]. In other words, infrastructure alone does not create public value unless accompanied by active user participation [83, 84].

This situation exemplifies the paradox of physical development, in which policy success is judged by the completion of construction projects rather than by behavioral change among users. The underutilization of facilities is therefore not a sign of achievement but of governance inefficiency [85, 86]. Field interviews indicating that passengers prefer “Illegal terminals” over the official facility reinforce this conclusion. The infrastructure planning failed to consider the behavioral dimension of urban mobility—namely, that Padang residents prioritize speed and practicality over regulatory compliance.

6.14 Institutional Limitations and Weak Interagency Coordination

From an institutional perspective, the duality of authority between the Land Transportation Management Agency of West Sumatra and the Padang City Transportation Agency creates jurisdictional overlap and weak enforcement. The effectiveness of multi-actor public governance depends on coordination, collaboration, and interdependence [87, 88]. In Anak Air's case, weak synergy has led to disjointed policy implementation, with planning handled by the central authority. At the same time, operational responsibilities fall to the local government, without a coherent coordination mechanism.

This demonstrates a deficiency in governance capacity, defined as the institution's ability to manage cross-sectoral and multi-level policy systems. The absence of an institutional coordination forum leads to fragmented decision-making, such as the development of physical infrastructure without route integration or feeder transportation planning. In effect, policy exists on paper but not in practice, reflecting a top-down administrative approach rather than a collaborative governance model.

6.15 Social Dynamics and User Behavior

Sociocultural factors strongly influence the Anak Air Terminal's underperformance. Padang residents prefer informal pick-up points and roadside access over formal terminals, illustrating a cognitive disparity between public policy and local mobility patterns. The effectiveness of transport policy depends on user perceptions and preferences—citizens define efficiency by convenience, not regulatory compliance [89, 90]. For them, the official terminal is rational only if it offers practical advantages in time, cost, and accessibility. Conceptually, this suggests that improving terminal governance requires a shift from structural to behavioral strategies. Public education campaigns, economic incentives for operators, and efficient feeder transport can reshape mobility behavior. Rather than enforcing top-down regulation, the government must design policies that nudge behavior toward formal transport systems, in line with behavioral economics principles.

6.16 Managerial and Human Resource Constraints

From the standpoint of public management theory, the Anak Air Terminal suffers from inadequate human resource capacity. Only two operational staff members manage supervision, service, and maintenance, which significantly limits oversight and field enforcement. Organizational effectiveness depends on the interplay among structure, process, and HR [91, 92]. The absence of regular staff training, digital management systems, or performance evaluation mechanisms further hampers effectiveness. The terminal's inability to adapt to the digital transformation of transportation services—such as online ticketing or ride-hailing integration—illustrates institutional inertia.

Conceptually, this highlights the need for digital governance transformation to improve responsiveness, transparency, and efficiency.

6.17 Economic and Fiscal Sustainability Dimensions

Financially, the terminal's low utilization has led to insufficient revenue from vehicle levies and kiosk rentals to cover operational costs. This places the terminal below the fiscal break-even point, signaling unsustainable financial management. Effective governance requires accountability, transparency, and fiscal sustainability [93]. The Anak Air Terminal's dependence on central government subsidies indicates a failure to achieve these standards. This also reflects a market failure in public service delivery, where the supply of transport facilities exceeds public demand. Corrective intervention—such as economic incentives, subsidy schemes, and regulatory enforcement—is thus essential. Introducing feeder services, subsidized entry for bus operators, and commercial partnerships could enhance both terminal viability and surrounding economic activity.

Overall, the Anak Air Terminal case demonstrates that the core problem of governance effectiveness lies not in infrastructure deficits but in misalignment between policy design and stakeholder behavior. Referring to the Good Governance framework, effective governance depends on five pillars: participation, accountability, transparency, responsiveness, and efficiency [50, 61]. Among these, the Anak Air Terminal performs relatively well in transparency (service information and reporting) but poorly in participation and responsiveness, indicating an imbalance in governance quality.

Importantly, these mechanisms do not imply that infrastructure, finance, security, and culture are irrelevant; rather, their influence is best understood as moderating factors for specific links in the causal chain. For example, improved terminal facilities and passenger amenities can strengthen the implementation effect on utilization by increasing the attractiveness of terminal-based operations. Still, they cannot fully offset weak enforcement expectations. Similarly, financial incentives can increase operator compliance by shifting the cost-benefit balance in favor of participation. Security and safety perceptions can further influence passenger willingness to use the terminal, thus moderating the final step from operator compliance to sustainable utilization. Framing these dimensions as moderators maintains their policy relevance while preventing the analysis from becoming a list of undifferentiated correlations.

Infrastructure-based approaches fail here because the primary constraint is not physical assets, but rather the institutional mechanisms that transform assets into coordinated service behavior. Implementation research consistently shows that policy effectiveness depends on credible enforcement, clear roles between agencies, and routines that reduce discretion and uncertainty in day-to-day operations. In this case, the terminal's role as a central mobility hub is compromised when institutional coordination is fragmented and inconsistent. Without shared operational protocols and harmonized enforcement routines, compliance remains optional in practice. Operators then face a predictable incentive structure: entering the terminal can incur dwell time losses and reduce passenger numbers, while avoiding the terminal often carries limited risk of sanctions. Under such conditions, infrastructure provision becomes necessary but not sufficient because the last mile of governance is behavioral compliance shaped by implementation forces, not by the presence of buildings and facilities.

In practice, the transportation system further explains why utilization remains low despite the facility's readiness: terminals are not stand-alone assets but rather nodes in a broader service network. Node performance depends on connectivity and on stable operational expectations for both operators and passengers. When connecting routes are limited or poorly integrated, passengers rationally prefer curbside pickup or alternative nodes that minimize transfer times and uncertainty. Low ridership then reduces the terminal's attractiveness to operators, reinforcing a negative feedback loop: weak connectivity and unreliable enforcement reduce utilization, and low utilization in turn weakens the incentives for operators to comply and for agencies to prioritize coordinated implementation. The result is a system-level equilibrium of underutilization that cannot be addressed by infrastructure improvements alone unless the governance and network integration conditions that generate utilization are also strengthened.

From an institutional analysis perspective, these findings highlight the gap between formal and implemented rules, which are underpinned by collective action and principal-agent dynamics. Many institutions hold partial authority, creating weakened accountability and opportunities for discretionary enforcement. In such situations, implementation on the ground resembles classic governance: the intensity of enforcement, the credibility of sanctions, and day-to-day coordination routines determine whether formal designs become effective practices. Operators respond to this institutional environment by choosing strategies that maximize revenue and minimize time costs, especially when the likelihood of meaningful sanctions is low. This explains why weak coordination and inadequate incentives are not merely descriptive labels but institutional mechanisms that shape compliance and utilization outcomes.

The broader contribution of this discussion is to demonstrate that the effectiveness of public transport services is generated through governance mechanisms, not just through infrastructure inputs. This case study indicates that effectiveness is best understood as the outcome of an institutional chain: coordination capacity shapes implementation power; implementation power shapes operator participation/compliance; and compliance determines utilization, which is the most proximate indicator of whether terminals function as effective public service nodes. This

relates to broader debates about governance and implementation by illustrating how fragmented authority and weak enforcement credibility can lock infrastructure projects into a low-performance equilibrium. In practice, the implication is that performance reforms should prioritize institutional coordination tools and arrangements that align with operator incentives, along with connectivity improvements that make terminal-based travel the least-friction option for users.

Beyond institutional fragmentation, the Anak Air Terminal case also illustrates how spatial and technical system constraints structurally influence governance outcomes. The terminal's remote location, narrow access routes, mixed traffic with local residential activities, and recurrent congestion, particularly during peak hours, create high costs for operators and passengers. Limited pedestrian infrastructure, the lack of dedicated public transport lanes, and weak signage further undermine adequate accessibility. In this context, the infrastructure-led approach underperforms not only because of poor inter-agency coordination, but also because the terminal as a network node has not yet been embedded in functional service patterns. Route design, feeder continuity, and timely connectivity remain weak, preventing operational reliability and passenger demand from co-evolving. Consequently, enforcement and administrative directives elicit predictable behavioral responses: operators rationally avoid entering the terminal when access delays and low passenger volumes increase operational risks, reinforcing informal pick-up points and perpetuating illegal terminals. Theoretically, the findings suggest that the effectiveness of terminal governance depends on the alignment of institutional rules and incentives with the spatial logic of network accessibility and performance that is, governance capacity should be combined with corridor-level traffic management, last-mile feeder provision, and accessibility enhancements for terminals to function as intermodal hubs rather than stand-alone facilities.

Beyond its single-case study design, this research offers analytical generalizations by clarifying transferable mechanisms through which infrastructure-led reforms can fail in multi-actor transport systems across many developing city contexts. The Padang case is treated as a "most likely" setting in which a formal terminal provision exists. Yet, utilization remains low because informal pick-up practices, weak feeder connectivity, and fragmented mandates create a stable equilibrium in which non-compliance is rational for operators and convenient for passengers. Under these conditions, physical completion does not translate into functional performance unless coordination routines, enforcement credibility, and incentive structures collectively shift behavior toward terminal-based operations. Therefore, its contribution is not a claim of statistical representativeness, but a portable causal explanation that can be tested across comparable terminals facing overlapping authorities and weak network integration. Future comparative research across different cities or terminals could test the boundary conditions and identify which configurations are sufficient to move the system from a low-utilization equilibrium to an effective intermodal node.

7 Conclusions

This study demonstrates that the effectiveness of the Anak Air Type A Terminal governance cannot be inferred solely from infrastructure provision. The evidence points to mechanisms by which fragmented institutional coordination undermines implementation, reflected in inconsistent monitoring routines, uneven enforcement credibility, and limited service integration. Weak implementation then reduces operator participation and compliance: operators rationally avoid entering the terminal when it incurs waiting time costs and limit passenger arrests, especially when sanctions are uncertain or non-binding. Low operator participation translates into persistently low terminal utilization, and because utilization is the most proximate indicator of whether a terminal is functioning as a public transportation node, overall governance effectiveness remains limited despite the availability of facilities.

Theoretically, these findings contribute to the broader debate on governance and policy implementation by clarifying why infrastructure-led reforms often fail in multi-actor public service settings. This case supports the institutional view of effectiveness: performance is generated not by physical assets themselves but by the coordination apparatus and enforcement arrangements that translate formal mandates into predictable routines. Where authority is fragmented, and enforcement discretion is high, compliance becomes optional in practice, and the system settles into a low-utilization equilibrium. Therefore, this study broadens the discussion of transportation governance by framing terminals as institutional nodes whose performance depends on aligned incentives, credible enforcement, and network-level integration, rather than on capital investment alone.

Methodologically, this study contributes to qualitative transportation research practice by making explicit the pathway from raw data to analytical claims. The analysis triangulates semi-structured interviews, structured observations of terminal operations, and a review of regulatory documents and operational records. A stepwise coding process is used to connect recurring patterns in the data with explanatory claims such as "weak coordination," inconsistent enforcement, and inadequate incentives. Furthermore, this manuscript introduces a concise operational bridge between theory and evidence by linking governance dimensions, utilization, enforcement/implementation, coordination, and connectivity with observable indicators and data sources. This strengthens transparency by clarifying what empirical signals support claims about effectiveness and where the evidence is convergent or limited.

Contextually, the case of Anak Air Terminal provides insight into transport governance in Indonesia's regional environment, where formal policy intentions and infrastructure investments coexist with entrenched informal pick-up practices and multi-level institutional fragmentation. These conditions are not unique to Padang and reflect challenges at many comparable terminals: overlapping mandates across agencies, weak feeder connectivity, and user preferences that prioritize convenience over formal compliance. While the findings are context-specific, the identified causal logic coordination shapes implementation, implementation shapes operator participation, and participation shapes utilization, offering a transferable explanation for why suboptimal utilization persists in similar multi-actor transport systems.

This study has limitations that also point to directions for future research. First, more robust administrative time-series data on enforcement actions and terminal utilization would enable more robust trend analysis and comparisons across terminals. Second, comparative studies across different terminals could test the boundary conditions of the proposed mechanisms, including market structure, route profitability, or variations in enforcement regimes. Third, a mixed-methods design can evaluate how incentive packages interact with enforcement credibility to change operator and passenger behavior over time. Practically, the policy relevance of this study follows directly from the mechanisms identified. Interventions should prioritize institutional solutions that make compliance rational and predictable: formalized interagency coordination routines, synchronized operational protocols, and consistent enforcement.

Author Contributions

Conceptualization, Z.A., A.H.R., R.Y., and F.E.; methodology, Z.A. and A.H.R.; validation, Z.A. and A.H.R.; formal analysis, Z.A. and A.H.R.; investigation, A.H.R.; resources, Z.A. and A.H.R.; data curation, Z.A., A.H.R., R.Y., and F.E.; writing—original draft preparation, Z.A. and A.H.R.; writing—review and editing, Z.A. and A.H.R.; visualization, Z.A. and A.H.R.; supervision, Z.A., A.H.R., R.Y., and F.E. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the findings of this study are available from the corresponding author upon request.

Acknowledgements

The research team appreciates and is most grateful to the Research Center for Policy, Governance, Development & Empowerment, Universitas Negeri Padang for conducting this research.

Conflicts of Interest

The authors declare that they have no conflicts of interest.

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